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The Relational Origins of the Conservative Online Media Niche, US Twitter 2022

Abstract

We investigate why conservative online news media are often seen as niche, whereas liberal outlets have ideologically broader audiences. We propose two explanatory mechanisms for this asymmetry. The behavioral explanation focuses on differences in homophily, where one ideological camp would be exposed to more cross-cutting content due to more diverse networking preferences. The structural explanation highlights how a platform's user base places some in the minority, naturally exposing them to more cross-cutting content. We analyze exposure and sharing of news media content among 420,000 US Twitter users in 2022, prior to Musk's acquisition of the platform. We find that conservative users, as the minority, were overexposed to cross-cutting media content, while liberal users, as the majority, were underexposed. Consequently, liberal media were shared across party lines, while conservative media were overlooked by liberals and circulated mostly within a tight network of conservative accounts. This paradox suggests that although conservatives primarily engage with their own media, liberal outlets attract a broader audience, including many conservatives. By combining observational data with simulated benchmarks, we find that the structural mechanism plays a primary role in the observed asymmetry, as exposure to liberal content extends farther into conservative online communities.

Keywords: online media, media fragmentation, homophily, curation bubbles, network diffusion, political polarization

1. Introduction

Why are liberal news media often perceived as mainstream, while conservative outlets are seen as niche (e.g., Nadler, Bauer, and Konieczna 2020; Mangold, Schoch, and Stier 2024; Pew Research Center 2025)? Using a previously untapped Twitter dataset on exposure to and sharing of online media content, we examine how the network structure of online communities contributes to highly asymmetric information ecosystems. Central to this phenomenon is the understanding that Twitter operates as a network, with following ties shaping who sees what content. We study the summer of 2022, the months leading up to Elon Musk's takeover of the platform in October of the same year. We find that conservative users—being in the minority—were overexposed to liberal-leaning content, while liberal users—who comprised the majority—were underexposed to conservative content. As a result, liberal-leaning media were shared across party lines, whereas conservative media were largely ignored by liberal users and circulated mainly within a tight core of strongly conservative accounts. This asymmetry in the dissemination of liberal versus conservative media content mirrors recent findings on Facebook (González-Bailón et al. 2023), which highlight "a substantial corner of the news ecosystem consumed exclusively by conservatives" (p. 392). The phenomenon is striking because it underscores an apparent paradox: while conservatives primarily engage with their own media, liberal outlets attract broader audiences, including many conservatives.

A natural explanation for the ideological asymmetry in the exposure to online news lies in different networking behaviors of conservatives and liberals. Users curate their information flow (and signal their identity) by following other accounts (Green et al. 2025), so that news stories reach them through network diffusion rather than having to search for headlines and articles. Differences in homophily rates when forming online ties offer a *behavioral* explanation for differential exposure to aligning and cross-cutting content. A body of literature portrays liberals as more tolerant to misaligned news content (Barberá et al. 2015; Bail et al. 2018; Chen et al. 2021) and more diverse in online tie formation (Conover et al. 2012; Boutyline and Willer 2017; Zimmerman et al. 2024). But there are also mixed results. Mosleh et al. (2021) show in a correspondence test involving artificial accounts fielded on Twitter that both liberal and conservative users, at roughly the same rates, were more likely to follow co-partisans. Bakshy, Messing, and Adamic (2015) report similar levels of network homophily among liberals and conservatives in observational data from US Facebook.

A second explanation of the asymmetry puzzle is *structural*. Twitter in 2022 provides an interesting testing ground: as a predominantly liberal platform, it placed conservative users in the minority within the networks that disseminate online news content. Being in the minority naturally exposes conservative users to more liberal accounts than vice versa. While members of the majority group can easily find and follow other ingroup accounts, the odds for ingroup interactions are against minority members. As the sociologist Peter Blau famously argued, under equal ingroup preferences, minority members inevitably form a larger share of outgroup ties due to an asymmetry in opportunity structure (Blau 1977; Blau and Schwartz 1984).[1] This pattern has been corroborated by numerous empirical studies across a diverse range of social settings, from classrooms and workplaces to marriage markets (Ibarra 1995; McPherson, Smith-Lovin, and Cook 2001; Arvidsson, Collet, and Hedström 2021; Puur, Rahnu, and Tammaru 2022). Due to this pattern, reduced homophily within the minority group broadens the ideological range of content that reaches them (for the link between homophily and exposure to novelty, see Granovetter 1973; Mark 2003; Aral and Van Alstyne 2011; Arvidsson, Hedström, and Keuschnigg 2025). The structural argument predicts that a minority (conservative users) is overexposed to cross-cutting content, whereas the majority (liberal users) is underexposed to cross-cutting content because they can be narrower networkers and thus stricter curators of their profile feeds. As a result, a surprising number of conservative-leaning users follow and engage with liberal news media.

To test the explanatory power of the *behavioral* and the *structural* accounts, we combine observational data with simulated benchmarks obtained under different assumptions of the network-generating mechanism. In our empirical analysis, we use Barberá et al.'s (2015) method to estimate the ideology of a sample of 420,000 US Twitter users (egos), their network contacts (alters), and the media outlets standing behind the URLs that appear in egos' profile feeds. Counterfactually rewiring ties among liberal and conservative users assuming different homophily preferences, we find, first, that substantial ingroup preferences are required to reproduce the observed asymmetric exposure pattern. Second, and most importantly, we find that networks formed under equal levels of homophily preferences among liberals and conservatives can reproduce the observed exposure asymmetry between the two groups. In other words, we do not need to make assumptions about differential behavioral tendencies in tie formation between ideological camps to reproduce the conservative media niche. Instead, we find strong evidence for the structural mechanism. But make no mistake, homophily preferences are required to bring about the phenomenon: without ingroup preferences, liberals and conservatives would have identical exposure to similar media diets. As such, behavioral tendencies also play a role—yet not in the hypothesized sense but as part of the structural account.

1. It should be noted that this argument assumes reciprocal ties and the same average degree in both groups. Minority members can break this pattern by either forming less ties overall, or by forming ties to a more concentrated set of ingroup members, producing greater indegree inequality in the minority group. But, under equal tie-formation tendencies, the minority group will necessarily form a greater share of outgroup ties.

2. Background

Online platforms are in constant flux, influenced by changes in user demographics, ownership, and the algorithms that moderate content and user experience (Salganik 2017; Hickey et al. 2023; Washington Post 2023; New York Times 2024; The Guardian 2025). Murthy (2024) offers an engaging summary of Twitter's history in a recent issue of the *Annual Review of Sociology*, describing its shift from a predominantly liberal platform after its launch in 2006 to a more conservative X in recent years. During the 2022 study period, 72% of US adults said they were using social media, 22% reported using Twitter, and only one-third of US respondents viewed social media as beneficial for democracy, while two-thirds saw it as harmful, and 79% believed it exacerbates polarization (Pew Research Center 2022, 2024).[2]

We study Twitter data from July and August 2022, a data window that co-occurs with a number of trending issues and events in US politics. The Biden administration was in its second year, with ongoing debates over Covid-19, election integrity, and voting rights. In February, Russia's invasion of Ukraine prompted widespread support for providing aid, before these issues became politicized. In June 2022, the Supreme Court overturned Roe v. Wade, sparking debates on reproductive rights and women's health. Throughout the year, there was a notable surge in gun violence, fueling discussions on gun control, public safety, and mental health. The Black Lives Matter movement remained relevant, though its prominence had diminished since its peak in 2020. The country grappled with significant economic challenges, including inflation rates not seen in decades, which contributed to heightened discussions on workers' rights, corporate responsibility, and economic inequality. In addition, issues like immigration, climate change, and the resurgence of Donald Trump dominated political discourse on Twitter during the summer of 2022.

3. Data and methods

We analyze a dataset of 420,000 US-based Twitter users from July and August 2022. These users are considered politically active as they follow accounts of high-profile politicians and pundits. Additionally, users were required to have posted at least one politically relevant URL during April and May 2022, the months leading up to the study period. The dataset was collected through the Twitter API, using the access points for users' timelines and lists of followers.

Our sampling strategy starts from a list of 962 politically relevant seed accounts including members of Congress and pundits identified from Wikipedia's list of US political commentators (https://en.wikipedia.org/wiki/Category:American_political_commentators). To identify active seed accounts on Twitter, we manually searched Twitter for profiles that we could reliably link to the names in the Congress and pundit lists. We prioritized accounts that appeared highest in search results and had at least 5,000 followers. In a second step, we sample from users who meet the following criteria: (1) they follow at least 3 of the seed accounts, (2) have a public Twitter profile, and (3) follow between 100 and 5,000 accounts. In the study period, only verified accounts could follow more than 5,000 accounts. By capping the collection of user accounts at 5,000 followers, we aim to capture a sample more likely to be individuals than organizations, and laypeople than celebrities, that better represents the general population of US Twitter users.[3] This procedure results in a sample of 418,590 users, who serve as the 'egos' in our analysis. We then collect egos' following networks, finding 44,128,479 unique 'alters' to whom the focal egos maintain ties. This results in ego-centered networks between approximately 420 thousand politically active Twitter users and the 44 million unique accounts they follow. To gauge the content shared between egos and their alters, we extract

2. Twitter/X data, like all social media data, are not representative of the general population. The Twitter/X user base in the US skews younger and more affluent than average (Blank 2017; Murthy 2024).

3. Focusing on users who follow at least three accounts that can be positioned on the ideological scale increases the precision of our classification. Relaxing this restriction would expand the sample by including users whose ideological leaning can only be inferred with greater uncertainty.

all posts that contain URLs from the 300 most popular domains in our dataset. Of the 300 most popular domains, we manually classified 138 domains (46%) as news media outlets.

We use Barberá et al.'s (2015) method to define the ideology of each of the focal users (egos), their network contacts (alters), and the media outlets standing behind the shared URLs of news stories which appear in egos' profile feeds. We estimate ideology scores in three steps. (1) We use a strict subset of seed accounts (members of Congress) and the most politically engaged egos who follow them to project their adjacency matrix into latent space (using singular value decomposition and correspondence analysis, as described in Barberá et al. 2015). This projection is taken to correspond to ideological position. (2) We project political pundits as secondary seed accounts onto the same latent dimensions. (3) We project the accounts of the remaining egos and alters onto the dimension that corresponds to political ideology. This measure of political ideology does not naturally correspond with traditional survey measures of political leaning and lacks intrinsic interpretability of absolute values. Instead, it places users on a relative continuum based on the political leaning of their social connections. At the boundaries of this continuum are those who only follow ingroup members located at the extremes in their respective ingroups. The ideology scale can further be used to quantify the political valence of a domain based on the ideological composition of its followers. By analyzing the ideological positions of its followers, we can project each media domain onto an ideological continuum. Our measure of domains' political valence strongly correlates with established external benchmarks: "R^2 = 0.83" with the AllSides media bias rating and "R^2 = 0.80" for the Ad Fontes political bias rating (see Appendix 1).

Table 1 describes the dataset from the perspective of users, classifying users' ideology relatively to the New York Times: users to its right are considered 'conservatives,' while those to the left are classified as 'liberals.' We choose the New York Times as a cut-point despite the fact that its perceived ideological position is context-dependent and contested, and external benchmarks classify the outlet as left-leaning (AllSides 2025; Ad Fontes Media 2025). We use it as a cautious cut-point—one that facilitates, rather than inhibits, the falsifiability of our theoretical argument. This decision errs on the side of classifying many moderates as conservatives, rather than narrowing the conservative category to only those on the far right. Even with this threshold, users categorized as conservatives comprise only 36% of the sampled ego accounts in the 2022 Twitter dataset. Among other things, Table 1 shows that liberal and conservative users were exposed to roughly the same volume of media URLs, on average 2,985 and 3,161, respectively (see the results section for further interpretation, and Appendix 2 for a re-run of our analysis using a cut-point further to the right).

Table 1. Description of the US Twitter dataset, July–August 2022.

	Liberal egos		Conservative egos	
	Number	%	Number	%
Accounts	263,102	64.3	145,855	35.7
Average ties	550.4		579.5	
Average ingroup ties	405.7	73.1	303.2	54.2
Ingroup alters' indegree inequality	0.853		0.850	
Average exposed media content	2985.1		3160.5	

We classify user accounts as 'liberal' or 'conservative' if their ideology is left or right of the New York Times position. We use the Gini coefficient to quantify indegree inequality among the alters that egos of the same ideology follow.

We combine these observational data with simulated benchmarks—obtained under different assumptions about liberals' and conservatives' homophily preferences—to explore the origins of the exposure asymmetry on Twitter. By simulation, we artificially intervene in users' ego-centered networks, manipulating what content reaches them. The key idea is that if exposure patterns in

the simulated networks align with those in the true network—while keeping homophily in tie formation constant—this would provide evidence supporting a structural explanation for how users are exposed to online news. We simulate ties (and thus exposure to media content) assuming identical ingroup preferences between liberal and conservative egos and investigate whether this still leads to ideological asymmetry in media exposure. If exposure asymmetry persists, it strengthens the case for the structural explanation behind the mainstream status of liberal media and the conservative media niche. To this end, we rewire the networks of egos following the methodology used in previous studies (Watts, Dodds, and Newman 2002; Centola 2015; Zhao and Garip 2021). For every ego i, the probability of forming a tie to alter j (x_{ij}) is given by $P(x_{ij}) = c \cdot e^{-\beta\gamma}$, where γ is the absolute ideological distance (in percentiles) between ego and alter, the parameter β regulates the strength of homophily preferences, and c is a normalizing constant. We set two counterfactual homophily conditions: $\beta = 0.005$ (weak homophily) and $\beta = 0.15$ (strong homophily). Under weak homophily, the likelihood of forming a tie decreases slowly and almost linearly with ideological distance, allowing for the formation of cross-cutting ties. The probability of forming a tie with an alter on ego's ideology percentile is only 1.1 times greater than with an alter 20 ranks away. By contrast, this factor is 20 under strong homophily. By comparing exposures in the simulated networks to those in the observed networks, we can infer whether differential levels of homophily explain the observed asymmetry in exposure.

4. Results

Figure 1 presents the ideology distribution of the 300 most popular domains shared by sampled Twitter users. Panel A focuses on the 138 news media domains among them, revealing a bimodal ideology distribution that highlights the polarization of news sources: the most shared content is dominated by clearly liberal and conservative domains, respectively, with conservative outlets clustering on the far right of the relative ideology scale. This level of polarization does not extend to other domains outside of news media, as shown in panel B, where domains such as Spotify, the Centers for Disease Control, and the scientific journal Nature are among the most widely shared. We still find bimodality for non-news-media domains, but there is no indication that any popular domains are exclusively shared by conservatives, as is evident in the long red tail of the ideology distribution for news media domains. In summary, polarization is a defining feature of news media on Twitter during the summer of 2022, but much less so of other domains.

In panel C of Figure 1, we observe the ideological asymmetry where liberal news domains are shared across the ideological spectrum, while conservative outlets remain niche. The panel reports the distribution of shares for specific domains across users' ideological ranks, where 1 represents the most liberal users, 100 represents the most conservative, and each percentile includes an equal number of users. This panel includes data for the four most liberal (blue) and four most conservative (red) news media domains. For contrast, we include the New York Times and Fox News, both of which, despite their ideological leaning, present a more moderate user composition. The asymmetry at the extremes is pronounced. Liberal domains are shared across a broad range of ideology ranks, including conservative users, as seen in the extended red right tails. By contrast, conservative domains are virtually never shared by liberal users as indicated by the narrow left tails. These domain-level snapshots present evidence of the conservative online media niche. The remaining analyses aim to explain this phenomenon, taking into account the fact that Twitter is organized as a network, where following ties shape content exposure.

Table 1 above describes the basic network features of our Twitter data, classifying users based on the ideological position of the New York Times: users to its right are considered 'conservatives,' while those to the left are classified as 'liberals.' Users classified as conservatives, even with this cautious cut-point, make up the minority (36%) of platform users. Despite following a roughly similar number of accounts on average (conservatives 580 and liberals 550), the share of outgroup

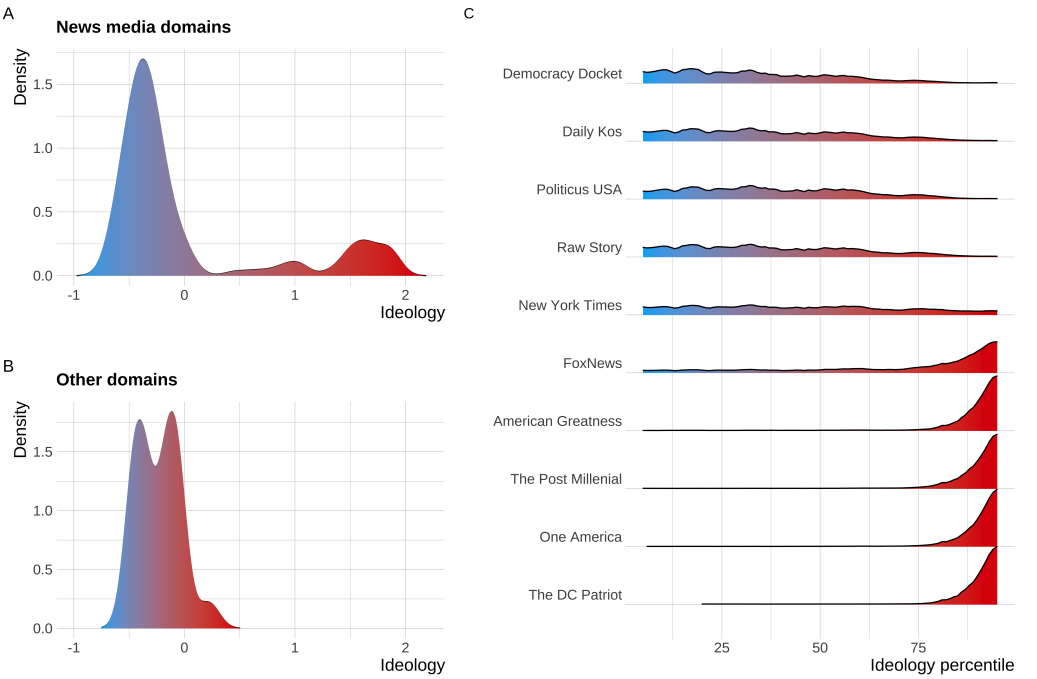

Figure 1. Twitter's conservative media niche: Distribution of ideological positions of (A) news media domains and (B) other domains shared July–August 2022. The projection of the ideology of domains considers shares of both ego and alter accounts. Ideology is relatively color coded, with blue indicating the outmost liberal domains in the sample, red for outmost conservative ones and a blend of red and blue for those in between. (C) Ideological distribution of the followers of the four most liberal (blue) and most conservative (red) news media domains as well as the New York Times and FoxNews as the most popular liberal and conservative news domains, respectively.

ties is substantially larger for conservatives (46%) compared to liberals (27%). At the same time, we find a similar concentration in followership—how evenly following ties are distributed among liberal and conservative accounts—as indicated by Gini coefficients of 0.850 for conservatives and 0.853 for liberals (where 0 would indicate perfect equality, with every alter receiving the same number of followers, and 1 indicating maximum inequality with all followers concentrated on a single account).

Distinguishing between what a focal user sees (exposures) and what they do (shares), Figure 2 shows sharing and exposure for the four most liberal and most conservative domains, as well as an average over the 10 most liberal or conservative domains, by egos' ideology percentiles. We use the ideology scale in terms of ranks (or percentiles), ranging from 1 to 100, with 1 being the most liberal and 100 the most conservative users, and each percentile including the same number of users. Comparing the plots in the top and middle rows, we observe a rather close correspondence between the sharing of political news content and exposure to such content through followed accounts. This result would seem to contrast with previous research (Bakshy, Messing, and Adamic 2015; González-Bailón et al. 2023; Robertson et al. 2023; Green et al. 2025), which has highlighted ideological filtering in the exposure-to-engagement funnel. Our domain-centric result shows some degree of filtering for the most liberal domains, as seen in the difference between exposures and sharings in the highest ideology percentiles (the most conservative users). Similarly, the red left tails are more pronounced for exposures than for sharings when looking at conservative media domains. But altogether, filtering is of a relatively lesser importance; even without it, the asymmetry prevails. Part of it is that liberals are not even exposed to the most conservative content, leaving them no chance to filter at this stage. Instead, the clearest and most consistent pattern in Figure 2 is that conservatives are exposed to the most liberal domains (heavy right tails in the blue plots), while liberals are not exposed to the most conservative domains (narrow left tails in the red plots). For example, if we look at the exposures to the most liberal domain Democracy Docket, focusing on the users above the 90th ideology percentile, we see that a meaningful proportion of these most conservative users' media exposures come from the most liberal outlet. By contrast, considering the most conservative domain, The DC Patriot, we see that virtually no liberals were exposed to it. These plots show a striking asymmetry not only in sharing behaviors but also in terms of exposures received. The central question then becomes: what accounts for this differential exposure; why are liberals less exposed to cross-cutting content than conservatives?

The *behavioral* explanation suggests that this asymmetry arises from different sorting preferences between conservatives and liberals. To maintain greater exposure to cross-cutting content, conservative Twitter users would need to choose broader networks than liberal users. The *structural* explanation, by contrast, attributes the exposure asymmetry to the minority status of conservative accounts within the Twitter network. The network statistics presented in Table 1 provide possible credence to both accounts. On the one hand, we do find that conservatives maintain broader networks than liberal users (46% of conservatives' alters are liberals, while only 27% of liberals' alters are conservatives). On the other hand, it is also the case that there are more liberal users on the platform (64% vs. 36%) and that, in support of the structural account, we do not find any evidence of the minority group creating fewer ties, or concentrating their ties on a more narrow set of ingroup users—which would be ways in which a minority could break the asymmetry in population structure.

To adjudicate between the behavioral and the structural accounts, we simulate counterfactual exposures under rewired social networks with different levels of homophily, examining their generative sufficiency. The bottom rows of Figure 2 compare the observed exposures to artificial exposures received in simulated networks. Following the procedure described above, we simulate counterfactual networks under different levels of ingroup preferences—but always keeping it the same for both groups. This produces networks of different levels of homophily. Under weak homophily (transparent histograms), users follow others across the ideological spectrum largely in proportion to their representation in the population, resulting in uniformly distributed exposures. As we in-

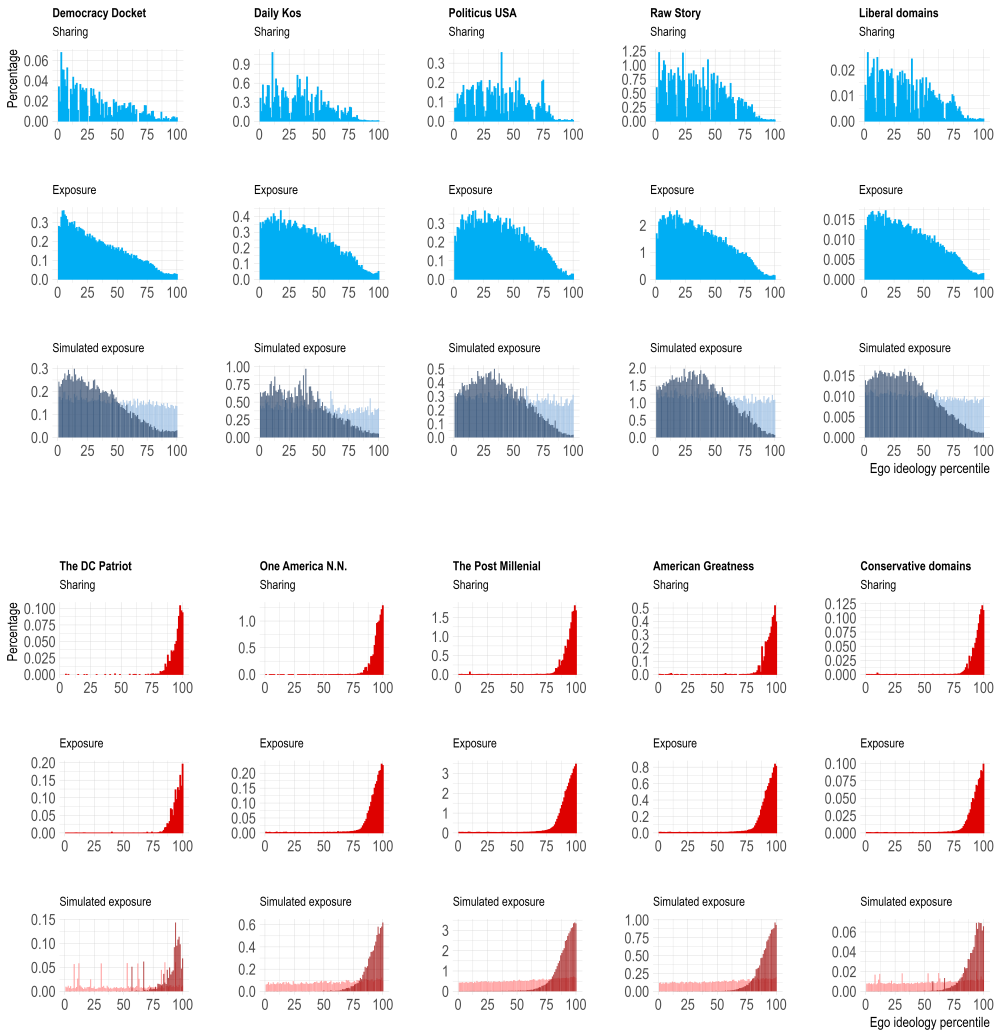

Figure 2. Exposure makes the difference: Distribution of egos' ideological positions who share (first rows) and are exposed to (second rows) content from the most liberal (blue) and most conservative (red) news media domains. On the y-axis, we plot the percentage of content users share and are exposed to against users' ideology percentiles (x-axis). The plots to the right show an average over the 10 most liberal or conservative news media domains, respectively. The bottom rows show simulated exposure benchmarks under assumptions of weak homophily (transparent) and strong homophily (opaque).

crease homophily, the simulated networks begin to approximate the true network in terms of the percentages of ingroup and outgroup ties for liberals and conservatives, such that simulated exposures increasingly resemble the observed ones. The simulation analysis demonstrates that assumptions about different homophily preferences between liberals and conservatives, in principle, are not required to reproduce the observed exposure asymmetry. Instead, the simulation strengthens the structural argument. Under negligible homophily in both camps, the asymmetry in exposure is lost—and with it the conservative online media niche. Under stronger levels homophily, both liberals and conservatives form more ties to ingroup members, but because of the asymmetry in the population structure, conservatives form relatively more ties to liberals (than liberals do to conservatives), making them overexposed to liberal content. Taken together, it is not how users of different ideologies form ties, but Twitter's ecosystem at the time that explains the heavy liberal and narrow conservative tails. As such, our results challenge a center-stage role of 'behavioral' explanations and support a structural explanation based on underlying population composition and network structure.

5. Discussion

To explain the ideological asymmetry in online media—where liberal content diffuses across party lines while conservative outlets remain confined to a far-right niche—we reported differences in exposure to cross-cutting media in the 2022 Twitter network. We found as the main mechanism behind the exposure asymmetry that, being in the majority, liberal users can afford to be narrow networkers, while conservatives—compelled into more outgroup ties—are more likely to be exposed to cross-cutting content. As a result, liberal-leaning media is shared across party lines, whereas liberal users overlook conservative media. Much like in schools where minority students form more outgroup friendships or in workplaces where individuals of the minority gender are compelled into outgroup interactions, our findings underscore that the asymmetry in media exposure on Twitter is driven primarily by structural constraints inherent to skewed group sizes, with in-group homophily playing a necessary yet secondary role in shaping these dynamics.

We have highlighted the structural and homophily-driven nature of the conservative media niche, yet social media algorithms may further modulate these dynamics. In the background of our observational data, a layer of algorithms already shapes how user feeds are organized, yet the extent of this influence is challenging to isolate empirically (e.g., Bakshy, Messing, and Adamic 2015; Salganik 2017; González-Bailón et al. 2023; Green et al. 2025). Recommendation systems, ranking algorithms, and content-personalization features can either reinforce majority–minority asymmetries (for instance, by preferentially surfacing ingroup posts) or mitigate them (by deliberately inserting cross-cutting content). This can interact with the minority status of certain ideological groups in subtle ways: a platform designed to promote 'variety' in user feeds might narrow the asymmetry by exposing majority users to more outgroup posts, whereas an algorithm that leans heavily on prior engagement patterns could deepen the existing exposure gap. Future research will connect to a growing body of work that examines how particular design choices—such as feed-ranking rules or link-recommendation systems—alleviate or amplify the structural forces that produce asymmetric information ecosystems (e.g., Karimi, Oliveira, and Strohmaier 2022; Neuhäuser et al. 2023).

Several limitations circumscribe our results. First, we approximate two political camps, but given the fractionalization in US politics there are definitely more. Second, we cannot rule out the possibility that conservative users follow and share liberal news media to critique or ridicule it—which could inspire future research efforts. At the same time, the patterns we observe could be driven by what some call the idea that "truth has a liberal bias," suggesting that liberal-leaning news stories, on average, line up more closely with facts (see, e.g., Garrett and Bond 2021). This could make liberal-leaning news more plausible, a feature that experimental research has shown to improve the ability of content to spread across ideological divides in partisan networks (Stein, Keuschnigg, and Rijt 2023). Third, we analyze posts of media content aggregated to the domain level.

Analyzing the sharing of individual news stories has been shown to more clearly flesh out ideological differences in news exposure and engagement (González-Bailón et al. 2023; Green et al. 2025). Still, the ideological discrepancy between the domain and the URL-level decreases as one nears the ideological end-points (Green et al. 2025), suggesting our results, that focus on these end-points, are likely to generalize to a URL-based analysis. Fourth, we have not given special attention to extreme users (for Twitter/X, see Baribi-Bartov et al. 2024). The distributions of ideology and activity are both skewed, such that it is reasonable to expect disproportionate influence by extreme accounts. An interesting avenue for future research is therefore to consider whether the described ideological asymmetry is sustained by a broad population of Twitter users or brought about mainly by clusters of extreme accounts. Finally, different conditions of dyadic influence will be empirically relevant. For example, closer alters will exert greater influence than more distant ones, and alters whose views are not perfectly aligned with ego's can increase exposure to novel ideas (Granovetter 1973; Mark 2003; Aral and Van Alstyne 2011; Arvidsson, Hedström, and Keuschnigg 2025). We also focused on first-order contacts and, of course, higher-order influences can be theorized, where the closeness of alters and the composition of the alters' networks influence egos' exposure to shared content. We invite future research to take up such investigations.

In retrospect, our analysis of 2022 Twitter captures a specific moment in time. A Pew survey conducted two years later, surveying X users posting about politics, found that 54% of conservative users cited feeling welcome as a major motivation, compared to 33% of liberals (Pew Research Center 2024). This trend is likely to have intensified, and conservatives, at this point, can no longer be expected to remain in the minority position they held during our observation period. Rather, if relative group sizes change, one would expect minority/majority (dis)advantages to shift, which in turn might cause the once-constrained group to retreat into newly available ingroup clusters, meaning declining attraction of liberal outlets among conservative audiences. While the concrete observations we analyze are historical, the mechanisms we study are general, making what we find in light of theory transportable to other platforms and periods. Shifts in group sizes rearrange exposure asymmetries in predictable ways. We encourage the use of multi-platform data—including emerging platforms like Bluesky or Threads—to validate whether similar structural constraints apply in contexts where the ideological majority differs or where platform features shape tie formation differently. In an evolving social media landscape, the fundamental structural insight remains: if an ideological group is small, it can rarely insulate itself fully and thus becomes disproportionately exposed to outgroup content.

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Competing Interests None.

Data Availability Statement The 2022 Twitter data and the code that support the findings of this study will be made available for download on a Github repository before publication. The replication package also includes sensitivity analyses and allows to conduct own robustness checks, alternative specifications of the computational analyses, and further exploration of the data.

Ethical Standards The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions All authors contributed equally to this research and are listed in alphabetical order. All authors approved the final submitted draft.

References

Ad Fontes Media. 2025. Media bias chart, https://app.adfontesmedia.com/chart/interactive?utm_source=adfontesmedia&utm_medium=website.

AllSides. 2025. Media bias ratings, https://www.allsides.com/media-bias/ratings.

Aral, Sinan, and Marshall Van Alstyne. 2011. The diversity-bandwidth trade-off. *American Journal of Sociology* 117 (1): 90–171.

Arvidsson, Martin, François Collet, and Peter Hedström. 2021. The trojan-horse mechanism: how networks reduce gender segregation. *Science Advances* 7 (16): eabf6730.

Arvidsson, Martin, Peter Hedström, and Marc Keuschnigg. 2025. Wide social influence and the emergence of the unexpected: an empirical test using Spotify data. *Sociological Science.*

Bail, Christopher A., Lisa P. Argyle, Taylor W. Brown, John P. Bumpus, Haohan Chen, M. B. Fallin Hunzaker, Jaemin Lee, Marcus Mann, Friedolin Merhout, and Alexander Volfovsky. 2018. Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences* 115 (37): 9216–9221.

Bakshy, Eytan, Solomon Messing, and Lada A. Adamic. 2015. Exposure to ideologically diverse news and opinion on Facebook. *Science* 348 (6239): 1130–1132.

Barberá, Pablo, John T. Jost, Jonathan Nagler, Joshua A. Tucker, and Richard Bonneau. 2015. Tweeting from left to right: Is online political communication more than an echo chamber? *Psychological Science* 26 (10): 1531–1542.

Baribi-Bartov, Sahar, John M. McDowell, Jane C. Anderson, Andrew J. Sullivan, Lucy D. Marks, Samuel K. Lee, Alexander T. Keller, George S. Chang, and Maria S. Foster. 2024. Supersharers of fake news on twitter. *Science* 384:979–982.

Blank, Grant. 2017. The digital divide among Twitter users and its implications for social research. *Social Science Computer Review* 35 (6): 679–697.

Blau, Peter, and Joseph Schwartz. 1984. *Crosscutting social circles: testing a macrostructural theory of intergroup relations.* New York: Routledge.

Blau, Peter M. 1977. A macrosociological theory of social structure. *American Journal of Sociology* 83 (1): 26–54.

Boutyline, Andrei, and Robb Willer. 2017. The social structure of political echo chambers: Variation in ideological homophily in online networks: Political echo chambers. *Political Psychology* 38 (3): 551–569.

Centola, Damon. 2015. The social origins of networks and diffusion. *American Journal of Sociology* 120 (5): 1295–1338.

Chen, Wen, Diogo Pacheco, Kai-Cheng Yang, and Filippo Menczer. 2021. Neutral bots probe political bias on social media. *Nature Communications* 12 (1): 5580.

Conover, Michael D., Bruno Gonçalves, Alessandro Flammini, and Filippo Menczer. 2012. Partisan asymmetries in online political activity. *EPJ Data Science* 1 (1): 6.

Garrett, R. Kelly, and Robert M. Bond. 2021. Conservatives' susceptibility to political misperceptions. *Science Advances* 7 (eabf1234).

González-Bailón, Sandra, David Lazer, Pablo Barberá, Meiqing Zhang, Hunt Allcott, Taylor Brown, Adriana Crespo-Tenorio, et al. 2023. Asymmetric ideological segregation in exposure to political news on Facebook. *Science* 381 (6656): 392–398.

Granovetter, Mark. 1973. The strength of weak ties. *American Journal of Sociology* 78 (6): 1360–1380.

Green, Jon, Stefan McCabe, Sarah Shugars, Hanyu Chwe, Luke Horgan, Shuyang Cao, and David Lazer. 2025. Curation bubbles. *American Political Science Review,* 1–19.

Hickey, Daniel, Matheus Schmitz, Daniel Fessler, Paul Smaldino, Goran Muric, and Keith Burghardt. 2023. Auditing Elon Musk's impact on hate speech and bots. *Proceedings of the International AAAI Conference on Web and Social Media* 17 (1): 1133–1137.

Ibarra, Herminia. 1995. Race, opportunity, and diversity of social circles in managerial networks. *Academy of Management Journal* 38 (3): 673–703.

Karimi, Fariba, Marcos Oliveira, and Markus Strohmaier. 2022. Minorities in networks and algorithms. *arXiv:2206.07113v1,* https://arxiv.org/pdf/2206.07113v1.

Mangold, Frank, David Schoch, and Sebastian Stier. 2024. Ideological self-selection in online news exposure: evidence from Europe and the US. *Science Advances* 10 (37): eadg9287.

Mark, Noah. 2003. Culture and competition: homophily and distancing explanations for cultural niches. *American Sociological Review* 68 (3): 319–345.

McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. Birds of a feather: homophily in social networks. *Annual Review of Sociology* 27:415–444.

Mosleh, Mohsen, Cameron Martel, Dean Eckles, and David G. Rand. 2021. Shared partisanship dramatically increases social tie formation in a Twitter field experiment. *Proceedings of the National Academy of Sciences* 118 (7): e2022761118.

Murthy, Dhiraj. 2024. Sociology of Twitter/X: trends, challenges, and future research directions. *Annual Review of Sociology* 50:169–190.

Nadler, Anthony, A.J. Bauer, and Magda Konieczna. 2020. Conservative newswork: a report on the values and practices of online journalists on the right. *Columbia Journalism Review,* March 31. https://www.cjr.org/tow_center_reports/conservative-newswork-report-on-the-values-and-practices-of-online-journalists-on-the-right.php.

Neuhäuser, Leonie, Fariba Karimi, Jan Bachmann, Markus Strohmaier, and Michael T. Schaub. 2023. Improving the visibility of minorities through network growth interventions. *Communications Physics* 6.

New York Times. 2024. How Elon Musk's own account dominates X (Nov. 3). https://www.nytimes.com/2024/11/03/technology/elon-musk-x-election.html?searchResultPosition=4.

Pew Research Center. 2022. Social media seen as mostly good for democracy across many nations, but US is a major outlier, https://www.pewresearch.org/global/2022/12/06/social-media-seen-as-mostly-good-for-democracy-across-many-nations-but-u-s-is-a-major-outlier/.

———. 2024. How X users view, experience the platform, https://www.pewresearch.org/internet/2024/06/12/how-x-users-view-experience-the-platform/.

———. 2025. The political gap in Americans' news sources: exploring use of and trust in 30 sources across U.S. political lines, https://www.pewresearch.org/journalism/2025/06/10/the-political-gap-in-americans-news-sources/.

Puur, Allan, Leen Rahnu, and Tiit Tammaru. 2022. What is the association between the ethnic composition of neighbourhoods, workplaces and schools and the formation of mixed-ethnic unions? *Population, Space and Place* 28 (1): e2504.

Robertson, Ronald E., Jon Green, Damian J. Ruck, Katherine Ognyanova, Christo Wilson, and David Lazer. 2023. Users choose to engage with more partisan news than they are exposed to on Google Search. *Nature* 618 (7964): 342–348.

Salganik, Matthew J. 2017. *Bit by bit: social research in the digital age.* Princeton: Princeton University Press.

Stein, Jonas, Marc Keuschnigg, and Arnout van de Rijt. 2023. Network segregation and the propagation of misinformation. *Scientific Reports* 13 (1): 917.

The Guardian. 2025. How Elon Musk's X became the global right's supercharged front page (Jan. 04). https://www.theguardian.com/technology/2025/jan/04/elon-musk-x-trump-far-right.

Washington Post. 2023. Elon Musk's X is throttling traffic to sites he dislikes (Aug. 16). https://www.washingtonpost.com/technology/2023/08/15/twitter-x-links-delayed/.

Watts, Duncan J., Peter S. Dodds, and Mark E. J. Newman. 2002. Identity and search in social networks. *Science* 296 (5571): 1302–1305.

Zhao, Linda, and Filiz Garip. 2021. Network diffusion under homophily and consolidation as a mechanism for social inequality. *Sociological Methods & Research* 50 (3): 1150–1185.

Zimmerman, Federico, David Beiley, Goran Muric, Emilio Ferrara, Jonas Schöne, Rob Willer, Eran Halperin, Joaquín Navajas, James Gross, and Amit Goldenberg. 2024. Attraction to politically extreme users on social media. *PNAS Nexus* 3:pgae395.

Appendix 1. Validation of domains' estimated political valence

Our follower-based measure of the ideological positions of news media domains maps well onto two established external benchmarks, the AllSides "Media Bias Rating" and the Ad Fontes "Political Bias Rating". Both rate U.S. online media content and reflect political bias rather than accuracy or credibility. AllSides ratings are derived from a combination of expert evaluations and blind surveys, in which participants assess content without knowledge of the source. The resulting political bias scale runs from -6 (left) to 6 (right). Ad Fontes uses trained analysts that rate articles prominently featured on the outlets' websites. Their bias scale runs from -42 (left) to 42 (right). Figure A1 speaks for the validity of the follower-based measure: the linear correlation is *X.XX* with the AllSides and *X.XX* with the Ad Fontes rating, respectively.

Appendix 2. Re-run with alternative liberal–conservative cut-point

In the main analysis, we have used the New York Times' ideological position as a cautious cut-point to classify users into conservatives or liberals. On the AllSides metric, the NYT leans left, with a rating of -2.2. We derive an alternative cut-point from the AllSides external ratings, choosing a large outlet in the center of their scale, The Wallstreet Journal (slightly conservative leaning with a rating of 0.33), as an alternative cut-point user classification. In Figures A2 and A3 we report our analysis based on this alternative left-right classification. Our results are robust to the choice of cut-point.